

# ROUND 1 STRATEGY FOR GUIDANCE SYSTEM SURVIVAL TESTING

10-day concept-validation roadmap for the hackathon submission round

Objective: show the judges a credible engineering pathway, early evidence, and clear progress - not a finished qualification prototype.

**UNDERSTAND THE RISK**

**DEFINE THE ARCHITECTURE**

**BUILD EARLY EVIDENCE**

**SHOW THE PATH TO QUALIFICATION**

# 1. Reframe the goal for Round 1

## ROUND 1 MISSION

Do not try to prove the complete system. Prove that the team understands the problem, has selected a technically meaningful risk, has designed a credible architecture around it, and has already started reducing uncertainty through analysis and early functional validation.

**Recommended positioning:** Before guidance accuracy can be improved, the embedded intelligence must reliably survive and remain healthy in a severe mechanical environment. Our approach is to develop a modular environmental-survivability and health-monitoring architecture, validate it digitally and through safe benchtop experiments, and then progressively qualify it for integration.

**The story judges should remember:**

|                                     |                                       |                         |                                      |
|-------------------------------------|---------------------------------------|-------------------------|--------------------------------------|
| We identified the foundational risk | We designed an architecture around it | We built early evidence | We have a path to full qualification |
|-------------------------------------|---------------------------------------|-------------------------|--------------------------------------|

# 2. What should exist 10 days from now

| Deliverable                       | What it proves                                                                     |
|-----------------------------------|------------------------------------------------------------------------------------|
| 1. System architecture            | You understand the complete challenge and where your selected workstream fits.     |
| 2. Failure / risk map             | You understand what can go wrong and why survivability matters.                    |
| 3. Conceptual mechanical CAD      | You have considered packaging, modularity and integration.                         |
| 4. Electronics architecture       | You know how sensing, health monitoring, logging and telemetry interact.           |
| 5. Small software / benchtop demo | You have moved beyond PowerPoint and generated early evidence.                     |
| 6. Validation roadmap             | You know how the concept progresses from functional proof to formal qualification. |

**The strongest Round 1 submission is not the one with the most subsystems. It is the one with the clearest reasoning and the best evidence of progress.**

### 3. Ten-day execution plan

#### DAY 1 - Decompose and narrow the problem

Break the challenge into major technical risks: guidance accuracy, navigation, actuation, electronics survivability, power reliability, communication, environmental robustness and safety. Select electronics survivability + health monitoring as the team's focused Round 1 wedge.

**Output: one problem-decomposition slide showing why this is a foundational risk.**

#### DAY 2 - Build the failure tree

Map system-level failure modes such as controller reset, software hang, sensor communication loss, corrupted readings, power interruption, communication loss and mechanical/electrical connection failure. For each, define detection logic, response concept and a safe validation method.

**Output: failure -> detection -> response -> validation matrix.**

#### DAY 3 - Freeze the concept architecture

Create the architecture that will appear consistently across the deck: environment -> rugged embedded module -> sensors/controller/power -> health monitor -> data logging + telemetry -> dashboard. Show future guidance/navigation/control/actuation as later layers.

**Output: final architecture diagram + interface view.**

#### DAY 4 - Create conceptual CAD

Develop an exploded view, cutaway view and a generic cylindrical housing concept. Show internal electronics zones, modular mounting, removable units and conceptual ruggedization features. Keep it clearly framed as demonstration packaging, not an operational design.

**Output: 2-3 CAD renders suitable for the deck.**

#### DAY 5 - Build the smallest proof-of-concept

Create a safe bench setup such as controller -> motion sensor -> laptop. Demonstrate system health, disturbance detection, sensor fault detection and recovery.

**Output: first working functional demo or recorded screen capture.**

#### DAY 6 - Build the dashboard

Show only decision-useful information: controller health, sensor health, power status, communications, logging, disturbance indication, event history and PASS / DEGRADED / FAIL.

**Output: polished live or recorded dashboard.**

#### DAY 7 - Run early functional tests

Perform baseline, controlled disturbance, sensor-disconnect and communication-interruption tests. Record expected versus observed behavior. Do not label these as qualification tests.

**Output: early validation table + screenshots / logs.**

#### DAY 8 - Build the validation ladder

Define the path from Level 1 functional bench validation -> Level 2 controlled laboratory testing -> Level 3 environmental qualification -> Level 4 integrated subsystem testing -> Level 5 full-system qualification by authorised facilities.

**Output: roadmap showing exactly where the team is today.**

#### DAY 9 - Build the competition story

Turn the technical work into a 10-slide narrative: problem, decomposition, insight, solution, architecture, CAD, progress, results, future roadmap, why this approach wins.

Output: near-final deck + demo flow.

DAY 10 - Attack your own submission

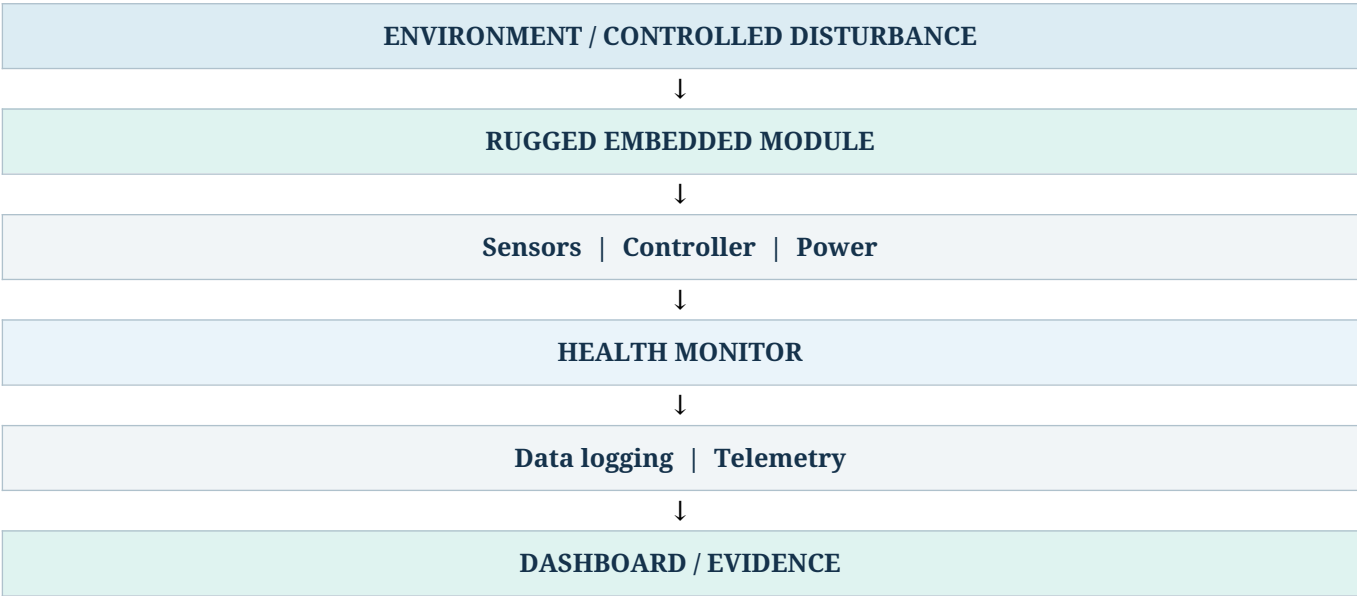
Stop building. Rehearse hard judge questions, tighten claims, verify evidence, remove overstatement and make every team member capable of explaining the architecture and current maturity level.

Output: final deck, demo recording / fallback, Q&A sheet and submission package.

4. Day 2 risk matrix - what judges should see

| Failure            | Detect using                | Response concept            | Safe early test                 |
|--------------------|-----------------------------|-----------------------------|---------------------------------|
| Sensor disconnect  | Health heartbeat / timeout  | Flag degraded state         | Disconnect the sensor           |
| Controller restart | Boot counter / restart log  | Log unexpected reset        | Deliberate software restart     |
| Power anomaly      | Voltage / status monitoring | Warning or degraded mode    | Benchtop power-fault simulation |
| Communication loss | Message timeout             | Log fault and status change | Pause telemetry stream          |

5. Architecture to keep consistent across the submission



Future layers: Navigation -> Guidance -> Control -> Actuation

IMPORTANT CLAIM BOUNDARY

The Round 1 demonstrator validates health monitoring, logging and fault detection logic under safe bench conditions. It does not claim to reproduce or qualify actual launch conditions. That distinction makes the submission more credible, not less.

6. Minimum viable proof-of-concept

- Controller connected to a motion sensor and laptop.
- Dashboard displays controller health, sensor health and communications status.

- Manual movement or safe controlled disturbance triggers a "disturbance detected" state.
- Disconnecting the sensor triggers a sensor-fault state.
- Reconnecting it triggers recovery and creates an event-log entry.

**Why this matters:** it changes the team from "we have an idea" to "we have started validating the architecture."

## 7. Early functional validation plan

| Test                          | Purpose                                     | What to monitor                         | Round 1 result format              |
|-------------------------------|---------------------------------------------|-----------------------------------------|------------------------------------|
| A. Baseline                   | Establish normal behaviour                  | health/status, logging, communications  | Stable baseline screenshot + log   |
| B. Controlled disturbance     | Show system remains observable under motion | sensor activity, resets, communications | Before / during / after comparison |
| C. Sensor fault               | Validate fault detection                    | timeout, fault flag, recovery           | Fault detected + recovered         |
| D. Communication interruption | Validate state awareness                    | telemetry loss and event logging        | Loss identified + logged           |

## 8. Validation ladder

| Stage   | Purpose                                            | Status      |
|---------|----------------------------------------------------|-------------|
| LEVEL 1 | Functional bench validation                        | WE ARE HERE |
| LEVEL 2 | Controlled laboratory mechanical testing           | NEXT        |
| LEVEL 3 | Environmental qualification testing                | LATER       |
| LEVEL 4 | Integrated subsystem testing                       | LATER       |
| LEVEL 5 | Full-system qualification by authorised facilities | FINAL       |

## 9. Recommended 10-slide competition story

| Slide | Title                 | Purpose                                                                             |
|-------|-----------------------|-------------------------------------------------------------------------------------|
| 1     | Problem               | Precision requires intelligence; intelligence requires survivability.               |
| 2     | Problem decomposition | Show all major engineering challenges and highlight the selected foundational risk. |
| 3     | Key insight           | A guidance algorithm with unavailable electronics has zero value.                   |

|    |                                      |                                                                                                           |
|----|--------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 4  | Our solution                         | Rugged modular embedded platform + continuous health monitoring + event logging + progressive validation. |
| 5  | How it works                         | Sense -> Monitor -> Detect -> Log -> Recover / degrade.                                                   |
| 6  | Mechanical concept                   | Exploded / cutaway conceptual CAD.                                                                        |
| 7  | What we have already built           | Controller + sensor + software + dashboard + first health checks.                                         |
| 8  | Early results                        | Functional validation test matrix and evidence.                                                           |
| 9  | How it becomes the complete solution | Today -> next integration layers -> laboratory validation -> qualification.                               |
| 10 | Why this approach wins               | Risk-first, modular, evidence-driven.                                                                     |

## 10. Judge questions you should rehearse

### "You have not tested actual launch conditions."

Correct. The current proof-of-concept demonstrates system-health monitoring and fault detection. Environmental qualification is a later stage, and we deliberately do not present benchtop testing as final qualification.

### "Why not focus on guidance accuracy?"

Guidance accuracy remains part of the full roadmap, but it depends on functioning navigation, control and electronics. We chose to de-risk a common enabling platform first.

### "What is innovative here?"

The architecture combines modular rugged electronics, continuous embedded health monitoring, event logging and an evidence-based qualification strategy so failures are detected rather than silently affecting downstream functions.

### "Where are you today?"

Architecture defined -> risk model completed -> packaging concept developed -> health-monitoring logic implemented -> early functional validation underway.

## 11. Recommended effort allocation

| Workstream                                                                                                                                                                                                                                           | Share of effort |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Problem / risk analysis                                                                                                                                                                                                                              | 15%             |
| Architecture                                                                                                                                                                                                                                         | 15%             |
| CAD                                                                                                                                                                                                                                                  | 15%             |
| Basic proof-of-concept                                                                                                                                                                                                                               | 25%             |
| Validation / results                                                                                                                                                                                                                                 | 15%             |
| Presentation / story                                                                                                                                                                                                                                 | 15%             |
| <b>DO NOT OVER-INVEST IN CAD</b><br>A beautiful render with no evidence is weaker than acceptable CAD plus a functioning health-monitoring demo and a disciplined validation story. The proof-of-concept should receive the largest share of effort. |                 |

## 12. Day-10 status checklist

|                                                              |                                                                |
|--------------------------------------------------------------|----------------------------------------------------------------|
| <input type="checkbox"/> Problem decomposed                  | <input type="checkbox"/> Critical failure modes identified     |
| <input type="checkbox"/> System architecture frozen          | <input type="checkbox"/> Packaging concept developed           |
| <input type="checkbox"/> Health-monitoring logic implemented | <input type="checkbox"/> Basic functional demonstrator running |
| <input type="checkbox"/> Early validation results available  | <input type="checkbox"/> Qualification roadmap defined         |

## 13. The key shift from the final-round roadmap

**Final-round goal:** Build and test an environmental-survivability prototype.

**Round-1 goal:** Show that the team has developed and started validating a credible engineering pathway toward environmental survivability.

### WHAT THE JUDGES ARE REALLY DECIDING

Do these people understand what they are building? Have they chosen a technically sensible approach? Can they execute it? Have they already generated meaningful evidence of progress? If the answer to all four is yes, Round 1 has done its job.